SEAGAS - Work Report: NLP Integration, Security Fixes and Project Status

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COS40005 Computing Technology Project A | Group 10 | POC004 | Sprint 2 report due 24 October

1. Summary

Today the NLP module (Ng Yong Hin) was connected to the backend, so both assessment flows now work end to end: choosing a role template, and pasting a job description. I then fixed the backend items assigned to me in the 16 October progress check: unauthorised reads of other students' records, the default secret key, password and input validation, and four functional bugs. Semantic and hybrid matching are installed and verified on my machine.

Status of this work: all changes are on branch `feature/nlp-backend-integration` and are **not yet committed or merged into dev**. Items marked * in section 4 become available to the team once that branch is merged.

Area	Result
NLP integration	Pasted JDs are analysed by `extract_skills()`; assessments use `match_skills()`; every skill result has a reason sentence, stored and shown in the UI.
Security (P0)	Students can no longer read other students' assessments, recommendations or JDs; no default secret key; password and input validation; no more 500 errors on bad input.
Bugs (P1)	Delete skill works; dashboard dimension scores load; experience scores follow added projects and certificates; advisor Gap % counts each student once.
Tests	85 end-to-end checks across three suites, all passing; 0 server errors (section 3).

2. Work Done Today

2.1 NLP integration (docs/NLP_INTERFACE.md, section 6)

Item	Change
(c) JD analysis	`POST /api/jobs/descriptions/{id}/analyse` calls `extract_skills()` with the skills_library rows and stores the results in `jd_extracted_skills`. It replaces the Sprint 1 placeholder, which marked JDs complete without extracting anything. Re-running replaces the previous results, so a JD can be compared across methods.
(c) Matching	`run_assessment()` uses `match_skills()` instead of substring matching (which treated Java as part of JavaScript and R as part of Docker).
(b) (d) Results stored	`matched_student_skill` and a new `reason` column are saved in `skill_match_results` and returned by `GET /api/assessments/{id}`.
(e) Canonical names	The JD path uses the library name ("Power BI") instead of the raw JD text ("PowerBI").
(f) Import path	New `backend/nlp_service.py`: every backend call into `nlp/` goes through it.
(g) Dimension scores	Developing now counts half in the 7 dimension scores (it counted 0).
(h) Default method	Hybrid, the same as the NLP module.
Section 8 warm-up	The model loads in a background thread when the backend starts, so the first student does not wait 20-30 s.

- **Fallback:** if the semantic packages are not installed, semantic and hybrid requests run as keyword and the student sees a short notice instead of an error. Teammates without the ~1 GB install can still use the site.
- **Frontend:** Job Analysis has a Hybrid / Keyword (Level 1) / Semantic (Level 2) choice, useful for the FR-14 comparison and the demo; each skill shows its reason; Skill Gap shows the reason on hover.

- **Installed:** `nlp/requirements.txt` on my machine (Python 3.12.5). On the brief's example JD, hybrid found SQL, Power BI, Data Visualisation, Critical Thinking and Communication.

2.2 Fixes from the 16 October progress check (section 5.1)

Priority	Issue	Fix
P0	Unauthorised reads	New `backend/access.py`: students only see their own assessments, recommendations and JDs; others get 404, the same as a missing record. Advisors and admins see all. Shared dataset JDs are readable by all but only staff can re-analyse them.
P0	Default secret key	The backend refuses to start without SECRET_KEY. `ensure_secret_key.py` (run by start_backend.bat) gives .env a random key and replaces the old public key that is in git. Users sign in again once.
P0	Passwords	One rule: 8 characters to 72 bytes, for register and admin. Long passwords now give 400 (register) or 401 (login) instead of 500 (bcrypt 5 raises above 72 bytes).
P0	Input validation	GPA 0-4, year 1-5, proficiency and project type limited to valid values, skill ID must be a UUID, text length limits: 422 with a message instead of a database 500.
P1	Delete skill did nothing	The skill list now returns `skill_id`, which the delete endpoint expects.
P1	Dashboard dimensions empty	`/assessments/my/history` returns the 7 dimension scores.
P1	Experience scores never changed	Profile counters are updated when a project or certificate is added; assessments count from the records (internship = larger of the profile field and internship projects).
P2	Advisor Gap % double-counted	`v_cohort_skill_gaps` uses only each student's latest assessment.

3. Testing

All backend tests ran against temporary databases that were deleted afterwards; `seagas_db` was not used for testing.

Suite	Result	What was checked
NLP, keyword only	22 / 22	Role and JD flows, reasons and matched skills stored, weighted score, fallback with notice, unknown method gives 400.
NLP with semantic	30 / 30	Brief example JD: keyword 4 skills, hybrid 5 (adds Critical Thinking), semantic 2; re-analysis replaces old rows. Speed: keyword 0.016 s, hybrid 0.048 s per JD.
Security and bugs	33 / 33	Student B cannot read A's assessment, recommendations or JD; advisor can; long and short passwords; GPA, year, proficiency, project type, UUID validation; delete skill; history scores; experience scores 75 to 100 and 66 to 99 after adding a project and certificate; Gap %.
Migration	Pass	A database built from dev's schema upgrades cleanly, also when run twice.
Frontend	Builds	Production build passes. Job Analysis was used in the browser (keyword path). A full walk-through of every page with hybrid is still to do.

4. Project Status Against the 13 Deliverables

Based on the 16 October progress check, updated with today's work. * = done on the integration branch, available after merge.

#	Deliverable	Status	Notes
1	Working web platform	**Mostly done***	Role template and pasted JD flows both work; single login with student, advisor and admin portals.
2	Student employability profile	**Partial**	Skills, projects, certificates; delete fixed*. Missing: evidence when adding a skill, work experience, extracurricular activities.
3	JD NLP module	**Done***	Keyword, semantic, hybrid; connected to the backend.
4	AI skill matching engine	**Done***	Strong / Developing / Gap with reasons, used by assessments.
5	Readiness score	**Done***	Weighted; developing counts half; experience dimensions update.
6	Skill gap analysis	**Done***	New matcher; reasons shown (hover on Skill Gap).
7	Recommendation engine	**Done**	Rule engine, 298 resources, beginner to advanced.
8	Student dashboard	**Done***	7 dimension scores now returned.
9	Advisor dashboard	**Partial**	Charts are nearly empty: synthetic students have no assessments (batch script needed). Report endpoint not used by the UI. "At Risk" wording to soften.
10	Development tracking	**Done**	Start / complete / summary.
11	AI evaluation results	**Partial**	Method comparison on 65 JDs done; 0 gold labels, so no precision / recall / F1 yet.
12	Usability evaluation (SUS)	**Planned**	Tasks and participants not prepared.
13	Documentation, report, demo	**Partial**	NLP and dataset documents done; README out of date.

Overall: 8 of 13 deliverables are done or mostly done once the branch is merged (2 on 16 October before today's work). The remaining gaps are evaluation evidence (11, 12), advisor data and oversight (9), profile completeness (2) and documentation (13).

5. What Is Still Missing

Owner	Task	Priority
Carl	Commit and merge `feature/nlp-backend-integration` into dev (pull request)	Now
Carl	Fix the git commit email (`@gnail.com`): GitHub does not count those commits	Now
Carl	Batch assessment script for the 5,000 synthetic students so the advisor dashboard has data (with Yong Hin)	P1
Carl	Audit log table: logins, registrations, advisor views of a student, assessments, admin changes	P1
Carl	Backend API tests with FastAPI TestClient (login, roles, access checks, assessment); turn today's test scripts into repo tests	P2
Carl	Update README (start-up steps, NLP install, tech stack)	P2
Yong Hin	Gold labels (20-30 JDs, second annotator for Cohen's kappa) for precision / recall / F1	P1
Yong Hin	Generic-word and alias fixes (e.g. "testing" to A/B Testing, "web-development" not matching React/HTML); rule proposals A / B / C with the team; TF-IDF baseline; 30 JDs per role	P1
Ren Hang, Shanice	Highlight extracted skills in the JD text; "decision support" notice on Dashboard and Skill Gap; "At Risk" to "Needs support"; show errors instead of empty catch blocks; evidence field when adding a skill	P1
Aaron	Human oversight: advisor review and notes on AI results; career path API and page	P1
All	SUS usability test; final report and demo	Sprint 4

6. Steps for Teammates After the Merge

- `git checkout dev` and `git pull`, then run `backend\start_backend.bat`.
- The database updates itself (reason column, Gap % view). The script replaces the old public secret key with a random one, so everyone signs in again once.
- Optional, for semantic and hybrid: `python -m pip install -r nlp\requirements.txt` (about 1 GB, mostly torch). Without it the site works with keyword matching and shows a notice.
- The first backend start after installing downloads the model (about 90 MB) and loads it in 20-30 s; the window then shows "NLP model loaded (hybrid ready)".